MODULE 14 · DOMAIN 3 - APPLICATION DEVELOPMENT

Augmenting prompts and shifting behaviour

Pulling key fields, terms and intents out of a message, and moving a model from its baseline to the answer you want.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 14 of 40

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A capable temp on their first day knows the language, not your customers. **Brief them with the facts that matter, and show them how you want it done.**

From a raw message to an augmented prompt

01
Message
The user's words, as typed.

02
Extract
Key fields, terms and intent.

03
Enrich
Look up records, filter retrieval.

04
Assemble
Fill a template with all of it.

05
Generate
The model answers a well-briefed prompt.

ON THE EXAM

The extracted fields are worth most when they change what is fetched, not just when they are pasted back in.

Fields, terms and intents, and what each one drives

EXTRACT	EXAMPLE	USE IT TO
Key fields	Order 4471, product KP-40, 12 March	Look up records, filter retrieval
Terms	Acronyms, jargon, product nicknames	Expand or normalise the search query
Intent	Complaint, how-to, billing, cancel	Route, or pick a template
Sentiment and urgency	Angry, frightened, time-critical	Set tone, or escalate
User attributes	Tier, region, language	Apply permissions and style

ON THE EXAM

Extraction can be a small model with structured output, or ai_extract and ai_classify in a pipeline.

A raw message, and the same message augmented

RAW

My KP-40 beeps. Covered?

- Product: unknown to the model
- Purchase date: unknown
- Warranty terms: not retrieved
- Model must guess

AUGMENTED

Intent: warranty check

- Product: KP-40 kettle
- Bought: 12 March 2025, from order lookup
- Warranty: 2 years, from policy retrieval
- Fault codes: KP-40 manual, filtered by product

ON THE EXAM

The augmented prompt answers the question before the model starts. The model's job becomes explaining, not guessing.

Paste the fields in, or use them to fetch?

PASTE IN ONLY

Fields as extra words

- Product code added to the prompt
- Retrieval still searches everything
- Wrong model's manual may come back
- Small benefit

USE TO FETCH

Fields change the context

- Product code filters retrieval
- Order number triggers a lookup
- Only relevant chunks arrive
- Large benefit

THE TIEBREAKER

When retrieval returns material for the wrong product, customer or period, extract the field and filter on it.

Baseline to desired: technique by gap

THE BASELINE IS	TECHNIQUE
Too long or too short	State the length, and show an example of it
The wrong tone or reading level	Name the audience, tone and level. Add examples
Inconsistent in format	Exact format spec, or structured output
Inventing facts	Answer only from context. Say so when unknown. Cite
Ignoring standing rules	Move the rules into the system prompt
Generic, not specific	Augment with fields and retrieved context

ON THE EXAM

Diagnose the gap between baseline and desired first. Each gap has its own technique.

Say what to do, or what not to do?

NEGATIVE INSTRUCTION

Do not be too technical

- Names the problem, not the target
- Leaves the model guessing
- Can prime the very thing forbidden
- Weaker on its own

POSITIVE INSTRUCTION

Two sentences, non-technical

- Names the target exactly
- Measurable: length and audience
- Easy to check in evaluation
- Stronger

THE TIEBREAKER

Describe the desired output. Keep negative constraints for hard limits, such as no personal data or no code fences.

Few-shot examples shift behaviour fastest

- Show **two or three answers written exactly as desired**: length, tone, structure.
- Vary the questions, keep the style constant.
- Show only good examples. Bad examples get copied too.
- Examples outweigh instructions. Make sure they agree.
- Put persistent tone and rules in the **system prompt**, examples alongside.

> ON THE EXAM
>
> When instructions alone do not move the tone, add examples before considering fine-tuning.

Iterate from baseline to desired

01
Baseline
Run the current prompt on a test set.

02
Name the gap
Length, tone, facts, format, rules?

03
One change
The technique for that gap.

04
Re-evaluate
Same test set, same measures.

05
Keep or revert
Fine-tune only if prompting fails.

ON THE EXAM

One change at a time, measured on the same set. Fine-tuning comes last, after prompting has clearly run out.

Changes that do not shift style or tone

- **A bigger context window.** More room, same voice.
- **A different embedding model.** Changes what is retrieved, not how it is written.
- **More retrieved chunks.** More material, same style.
- **A re-ranker.** Better order of context, same tone.
- **Temperature** changes variability. It does not make answers simpler or kinder.

> ON THE EXAM
>
> In a NOT question about style, the odd one out is usually a retrieval or capacity setting.

Pennant Outdoor's support prompt, before and after

01
Before
Long, technical, sometimes invents specs.

›

02
Augment
Product code filters the manual. Order lookup adds dates.

›

03
Ground
Answer only from context. Cite the manual section.

›

04
Style
Three sentences, plain English, two examples.

›

05
After
Short, specific, sourced, friendly.

ON THE EXAM

Each change fixed one named gap. Together they turned a generic, risky answer into a specific, safe one.

> A product support assistant retrieves from manuals for hundreds of appliance models. Users usually mention a model code, such as KP-40, in their message.
>
> Answers often quote the manual of a similar but different model.

What is the most effective change?

A Use larger chunks so each carries more of its manual

B Raise the temperature so the model considers more options

C Extract the model code from the message and filter retrieval on product metadata

D Add a system prompt telling the model to focus on the right product

◆ PAUSE AND DECIDE

What is the most effective change?

A Larger chunks

Similar manuals still look alike. Blurs vectors further.

B Higher temperature

More randomness, not better targeting.

C Extract code, filter retrieval

Uses the key field to fetch only that model's manual.

D System prompt to focus

The wrong manual has already arrived. Asking cannot fix that.

THE SIGNAL

Users mention a model code, and the wrong model's manual comes back. The key field is in the message: extract it and filter.

> An HR assistant retrieves relevant policy sections for each question. Evaluation shows answers sometimes state entitlements that appear nowhere in the retrieved sections.
>
> The team wants answers that stay within the policy text.

Which prompt change moves the baseline towards the desired output?

A. Answer only from the context, say when unknown, and cite it

B. Raise the temperature to encourage more careful answers

C. Remove the retrieved sections so there is less to confuse it

D. Ask for longer, more detailed answers with full explanations

◆ PAUSE AND DECIDE

Which prompt change moves the baseline towards the desired output?

A Answer only from context, cite

Grounds the answer and gives it a permitted way to not know.

B Higher temperature

More variability tends to mean more invention, not less.

C Remove the context

Leaves the model nothing to answer from but memory.

D Longer answers

More room for unsupported claims.

THE SIGNAL

Entitlements that appear nowhere in the retrieved sections. The gap is invention, and the technique is grounding with a way out.

> A bank's assistant gives accurate answers, but they are full of technical jargon. The bank wants short, plain-English answers suitable for any customer.
>
> The team is reviewing possible changes.

Which change would NOT help shift the responses to plain language?

A. State the audience and reading level, and a maximum length

B. Add two or three example answers written in plain English

C. Set a plain-English tone in the system prompt

D. Switch to an embedding model with more dimensions

◆ PAUSE AND DECIDE

Which change would NOT help shift the responses to plain language?

A Audience, level, length

Helps. A positive, measurable target.

B Plain-English examples

Helps. Examples shift style fastest.

C Tone in system prompt

Helps. A standing rule for every turn.

D More embedding dimensions

Does not help. Changes retrieval, not how answers are written.

THE SIGNAL

In a NOT question about style, find the retrieval or capacity setting. It cannot change how the answer is written.

A utility's assistant handles complaints, billing questions and moving-home requests with one general prompt. Complaints get curt replies, billing answers lack the customer's figures, and moving requests miss the new address.

The team wants each kind of request handled well.

Which approach best augments the prompts?

A Add every rule for all three request types to one prompt

B Classify intent, extract key fields, fill a matching template

C Fine-tune the model on past replies of all three request kinds

D Use a model with a larger context window for all the rules

◆ PAUSE AND DECIDE

Which approach best augments the prompts?

A One prompt, every rule

Longer and more confusing. Still no customer figures.

B Intent, fields, template

Each type gets its tone, its lookup and its fields.

C Fine-tune on past replies

Cannot supply this customer's figures or new address.

D Larger context window

Room for more, but nothing new is fetched.

THE SIGNAL

Three request types, each missing something different. Intent picks the template; key fields fill it.

What to carry forward

- Extract **key fields, terms and intents**. Use them to fetch, not just to paste.
- Wrong product, customer or period retrieved: **extract the field and filter**.
- Name the gap from baseline to desired, then apply its technique.
- Positive instructions and good examples. Ground with answer-only-from-context and citations.
- Retrieval and capacity settings do not change style. Temperature changes variability.

> ON THE EXAM
>
> Next: module 15 compares LangChain, LangGraph and the other frameworks for building these applications.